1. Write down the drawback of IRR. A person invests a sum of Rs. 2,00,000 in a business and receives equal net revenue of Rs. 50,000 for next 10 years. At the end of 10th year, the salvage value of the business is Rs. 25,000. Find the ERR of the business with έ=5%.
2. In Hanumaanghat Bhaktapur, the vehicle users take a roundabout to reach certain places because of the presence of a Hanumaante river. The result is excessive travel time and increased fuel cost. So the Bhaktapur Municipality is planning to construct a bridge across the river. The estimated initial investment for constructing the bridge is Rs. 50,00,000. The estimated life of the bridge is 15 years. The annual operation and maintenance cost is Rs. 1,60,000. The value of fuel savings due to the construction of the bridge is Rs. 6,00,000 in the first year and it increases by Rs. 40,000 every year thereafter till the end of the life of the bridge. Check whether the project is justified based on modified B/C ratio using AW formulation. Assuming an interest rate of 12% per year.
3. You are considering two investment options, In option A, you have to invest $ 5,000 now and $ 1000 three years from now. In option B, you have to invest $ 3,500 now, $1,500 a year from now and $ 1000 three years from now. In both options, you will receive four annual payments of $ 2,000 each. Which of these option would you choose based on a) the conventional payment criterion and b) the present worth criterion assuming 10% interest?
4. Auto numeric company has just brought a new spindle machine at a cost of $ 1,05,000 to replace one that had a salvage value of $ 20,000. The projected annual after tax saving via improved efficiency, which will exceed the investment cost, are as follows. What is the conventional payback period?

5. We have three alternative highway improvement to consider. In which should we choose to invest? All will last for 50 years. Assume an interest rate of 3%.

		Saving			
S.N	Construction cost	Accidents	Travel Time	Operation	O & M
Alternative A	$1,85,000	$5,000	$3,000	$ 500	$ 1,500
Alternative B	$ 2,20,000	$ 5,000	$6,500	$500	$ 2,500
Alternative C	$ 3,10,000	$ 7,000	$6,000	$2,800	$ 3,000

By B/C ratio and IRR method.

6. One of the two production units described below must be purchased. The MARR is 12%. Compare the two units. Which should be chosen based upon these economics?

	Unit A	Unit B
Initial cost	16,000	30,000
Life	8 years	15 years
Salvage Value	2,000	5,000
Annual operating cost	2,000	1,000

7. Given two mutually exclusive alternative, which one should be implemented? MARR 15%

Machine	A	B
Investment cost	-20,000	-30,000
Annual Revenues	10,000	14,000
Annual Expenses	-4,400	-8,600
Market value	4,000	0
Useful life	5 years	10 years

8. Select the best proposal among the mutually exclusive alternatives given below, using the internal rate of return method. The MARR = 6%

Years	A1	A2	A3
0	-200	-350	-300
1	50	100	70
2	50	90	70
3	50	80	70
4	50	70	70
5	50	60	70

9. One of the two production units described below must be purchased. The MARR is 12%. Compare the two units. Which should be chosen based upon these economics?

	Unit A	Unit B
Initial cost	16,000	30,000
Life	8 years	15 years
Salvage Value	2,000	5,000
Annual operating	2,000	1,000

10. A Contractor is considering buying a crane for his next bridge project. Using a Rate of Return analysis, with MARR of 7%, determine which alternative should be selected.

	Crane A	Crane B
Initial Cost	$9400	$5200
Uniform Annual Benefit	$1750	$1650
Useful life, year	8	4

11. A company is considering investing in a water purification system. Three alternatives are under consideration and their cash flow profiles are given below. The MARR = 10%. Select the best proposal among there alternatives, using benefit cast ratio method.

Years	System 1	System 2	System 3
0	-80,000	-120,000	-100,000
1	40,000	60,000	35,000
2	35,000	50,000	35,000
3	30,000	40,000	35,000
4	25,000	30,000	35,000

12. A Engineering firm is considering the following mutually exclusive alternatives. Which project would you select based on IRR method, assuming MARR = 9% per year for years of 10.

	Alternative A	Alternative B	Alternative C
First cost	-4,00,000	-3,60,000	-7,25,000
Net annual	1,10,000	79,000	1,66,000
Salvage Value	34,000	28,600	53,000

13. Three alternatives are being considered for improving an operation on the assembly line. Each of plans A, B and C has a 10 year life and a scrap value equal to 10% of its original cost. If interest is 8%, which plan should be purchase.

	Plan A	Plan B	Plan C
Installed cost	$ 15,000	$ 25,000	$ 33,000
Material and	$14,000	$ 9,000	$14,000
Annual	$ 8,000	$ 6,000	$ 6,000

14. Two pumps are being considered for purchase. If interest is 7% which pump should be brought?

	Pump A	Pump B
Initial Cost	$7,000	$5,000
Salvage value	$1500	$1000
Useful life in years	12	6

15. Consider these mutually exclusive alternative MARR = 8% per year. So all alternatives are acceptable.

	Alternative	Alternative B	Alternative C
Capital investment	$ 250	$375	$500
Uniform annual	$ 40.69	$ 44.05	$ 131.9
Useful life in years	10	20	5
		(* in Thousands)	

- At the end of their useful lives, alternatives A and C will be replaced with identical replacements. Which alternative should be chosen and why.

16. The manager is canned food processing plant is trying to decide between two labeling machines. Their respective costs and benefits are as follows:

	Machine A	Machine B
First Costs	$ 15,000	$25,000
O&M costs	$ 1,600	$ 400
Annual benefits	$ 8,000	$ 13,000
Salvage value	$ 3,000	$ 6,000
Useful life in years	7	10

Assume an interest rate of 12%. Use annual cash flow analysis to determine which should be selected.

17. Two machines are being considered for purchase. If the MARR is 10%, which machine should be brought? Use an IRR analysis comparison.

	Machine X	Machine Y
Initial Cost	200	700
Uniform annual benefit	95	120
Salvage value	50	150
Useful life in years	6	12

18. Which Alternative should select if study period is 6 years and MARR = 10%.

	A	B
Capital investment	3500	5000
Annual cash flow	1255	1480
Useful life (years)	4	6

19. Assume you are called on to maintain a cementry site forever if the interest rate = 4% and $50/year is required to maintain the site. How much amount should be deposited now to run that project forever.

20. Suppose we buy a machine for $ 50,000 and it has a useful life of 8 years. The market value at the end of 8 years is expected to be $10000. Study period is only 5 years and you need to estimate its market value at the point in time. MARR = 10%

21. Co-terminated assumption with an 4 years study period MARR = 10%

	A	B
Capital Investment	3500	5000
Annual cash flow	1255	1480
Useful life years	4	6
Salvage Value	1000	1000

22. Using co-terminated assumption recommend the best project taking study period as 8 years. MARR = 10%

Project	**A**	**B**
Initial Investment	3,50,000	5,00,000
Annual Revenues	1,30,000	1,75,000
Annual Cost	15,000	25,000
Salvage Value	35,000	50,000
Useful life	5 years	8 years

23. Using co-terminated assumption recommend the best project taking study period as 6 years. MARR = 10%

Project	**A**	**B**
Initial Investment	3,50,000	5,00,000
Annual Revenues	1,30,000	1,75,000
Annual Cost	15,000	25,000
Salvage Value	35,000	50,000
Useful life	5 years	8 years

24. Given the following independent projects, determine which should be chosen using the AW method. MARR = 10% and there is no limitation of fund availability. Each project has useful life of 5 years.

Project	First cost	Net Annual cash flow	Salvage value
X	10,000	2300	10,000
Y	12,000	2500	1000
Z	15,000	4000	1000

25. Consider the following mutually exclusive alternatives. Which alternative should be chosen if chosen alternative is required for infinite period? Take MARR = 10%.

Investments considered	Alternatives (Rs.'000)	
	A	B
Initial investment (I)	$15,000	$30,000
Annual Expenses (E)	$2,000	$1,000
Salvage Value (S)	0	$3,000
Useful life (N)	10 Years	60 Years